KRISHNAIK ACADEMY

AI FDE

Forward Deployed Engineer Bootcamp

V1.0

ENTERPRISE AI INTEGRATION

AGENTIC AI

RETRIEVAL AUGUMENT GENERATION

AI SECURITY & GOVERNANCE

MULTI-AGENT ORCHESTRATION

GENERATIVE AI

This intensive bootcamp transforms you into a AI Forward Deployed Engineer (AI-FDE). You will master core software engineering, serverless cloud deployment, and advanced Applied AI, including hybrid RAG, Knowledge Graphs, and Vision AI. Crucially, you will learn to orchestrate autonomous AI agents using LangGraph, securely integrate them with legacy enterprise systems (SQL, SOAP), and enforce strict corporate security (SSO, RBAC). By the end, you will possess both the technical execution and consulting skills needed to deploy cutting-edge AI into complex, highly regulated environments.

Learning Objectives

- Write scalable, asynchronous Python code and build production-ready REST and GraphQL APIs using FastAPI.
- Provision secure, cost-controlled AWS infrastructure (VPCs, IAM, RDS) and deploy serverless containers via Docker and ECS Fargate.
- Design and optimize Advanced RAG pipelines using vector databases (Pinecone, Qdrant), hybrid search (RRF), and semantic chunking.
- Model and query enterprise Knowledge Graphs using Neo4j and Cypher to power complex AI data traversal.
- Process unstructured enterprise data natively, utilizing Vision Language Models and OCR-free document parsing (ColPali).
- Architect multi-agent systems using LangGraph to execute complex reasoning, parallel tasks, and human-in-the-loop workflows.
- Securely connect AI agents to internal enterprise tools (Jira, Slack) using the Model Context Protocol (MCP).
- Integrate modern AI with legacy corporate infrastructure, navigating SOAP APIs, XML parsing, and on-premise SQL databases.
- Implement enterprise-grade Identity and Access Management (IAM), including OAuth 2.0, SSO, and data-level Role-Based Access Control (RBAC).
- Protect AI applications against prompt injection and data leakage using NVIDIA NeMo Guardrails and Microsoft Presidio PII masking.
- Monitor, evaluate, and trace LLM applications in production using API gateways and observability tools (LangSmith, Langfuse).
- Navigate the full enterprise consulting lifecycle—from leading technical discovery workshops and drafting architecture SOWs to delivering executive ROI presentations.

Course Information

Prerequisites

Basic **programming knowledge** (Python, SQL, APIs, and Git) and a **strong problem-solving** mindset are required. We start with foundational coding and debugging practices, so no prior experience with GenAI, RAG, Agents, Cloud, or DevOps is necessary. A willingness to learn and a drive to tackle complex, real-world enterprise problems are essential.

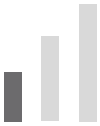
Approximately 4.5 to 5 months (18–20 weeks). This includes dense technical modules, live debugging, and two massive end-to-end enterprise capstone projects simulating real-world client engagements

Estimated Time



4-5 months 8hrs/week*

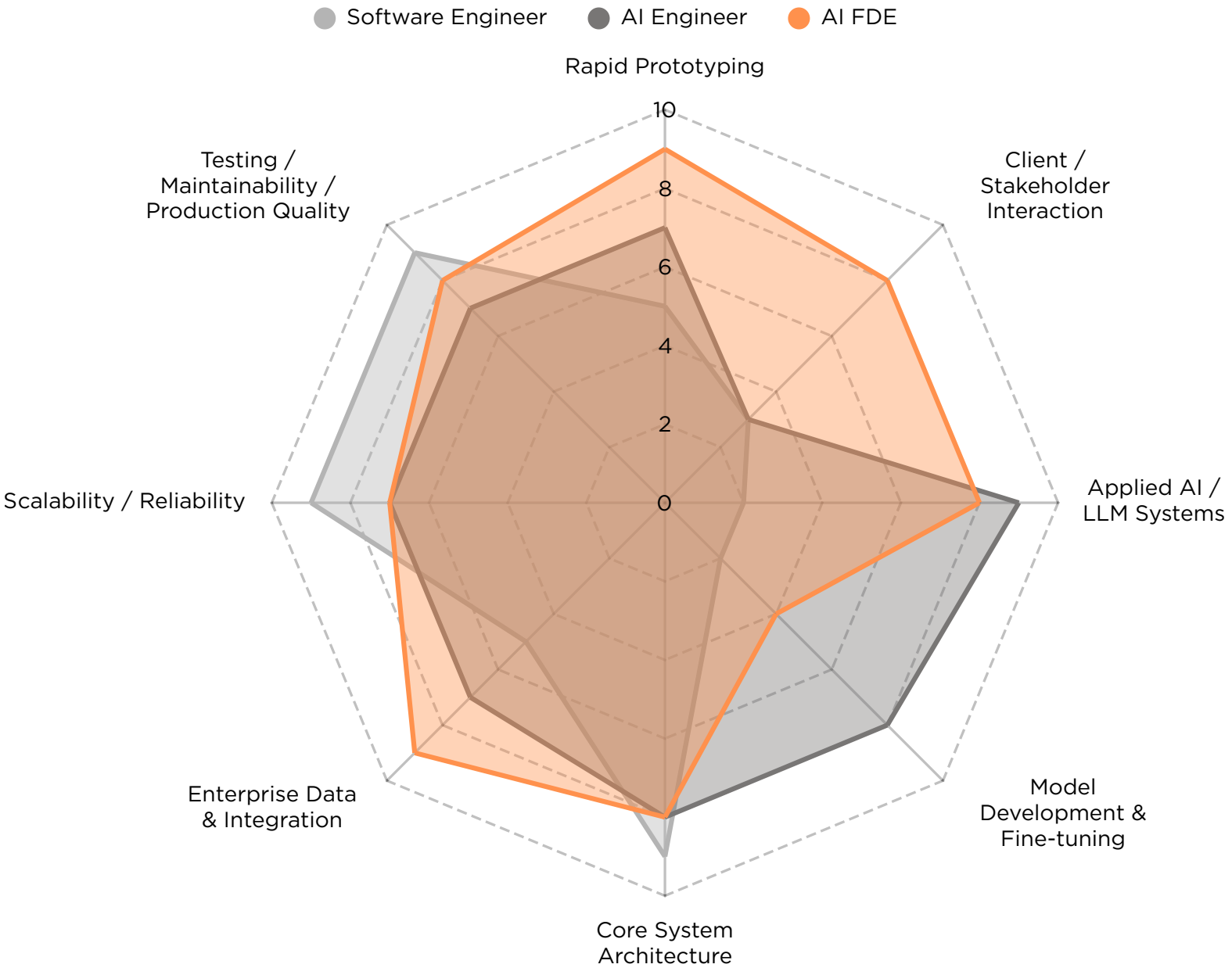
Required Skill Level



Begineer

Role Focus Matrix

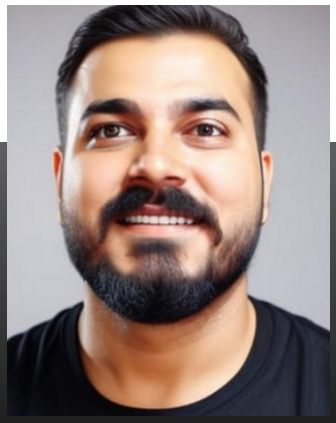
SDE vs AI-Engineer vs AI-FDE



Krish Naik Academy Team



Boktiar Ahmed Bappy
AI Engineer
Linkedin



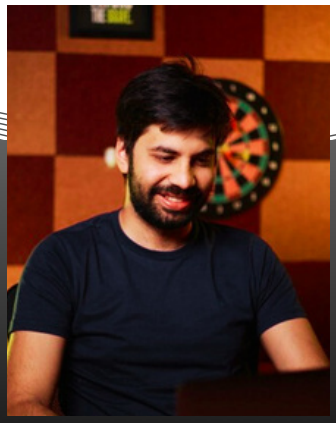
Krish Naik
Chief AI Engineer
Linkedin



Monal Kumar
Senior Data Scientist
Linkedin



Sunny Savita
GenAI Engineer
Linkedin



Mayank Aggrawal
Senior ML Engineer
Linkedin



Sourangshu Pal
Senior Data Scientist
Linkedin

Module 1

Python & Linux Foundations

In this module, learners establish the core programming baseline required for all subsequent engineering tasks. You will transition from basic scripting to advanced Python execution, focusing on memory management and asynchronous programming to handle high-performance workloads, while navigating enterprise Linux environments with ease.

Topics	
Python	Core data structures, Memory management fundamentals, Object-oriented programming basics, Exception handling, File I/O operations
Advanced Python & Async Programming	Event loop architecture, Coroutines and tasks, Async context managers, Concurrency vs parallelism, ThreadPoolExecutor integration
Linux for Engineers	Navigating the file system, Permission and user management, Process monitoring, Shell scripting basics, Environment variable configurations

Module 2

Modern API Development

This module focuses entirely on backend architecture. You will learn to expose your Python logic to the web by building scalable REST and GraphQL APIs. A heavy emphasis is placed on rigorous testing frameworks and developer productivity tools to ensure the code you write is production-ready.

Topics	
FastAPI & REST APIs	Path and query parameters, Pydantic data validation, Dependency injection, CORS middleware, Building scalable CRUD endpoints
GraphQL Basics & API Versioning	Designing GraphQL schemas and object types using Python frameworks like Strawberry or Graphene, Writing efficient data resolvers, Handling N+1 query problems, Executing queries vs mutations, Implementing URL-based vs Header-based API versioning
Testing (Pytest) & VS Code Productivity	Unit testing fundamentals, Fixtures and mocking, Test coverage analysis, Debugging configurations, Essential IDE extensions

Module 3

Cloud Fundamentals & Networking

Transition your architecture into the cloud. This module covers the foundational AWS infrastructure required to host enterprise applications. You will learn to provision compute and storage, secure your network boundaries with virtual private clouds, and enforce strict financial budgeting for cloud resources.

Topics	
AWS Fundamentals	Provisioning virtual machines, Object storage lifecycle policies, Managed relational databases setup, Event-driven serverless function basics
Networking Basics	Creating virtual private clouds, Public vs private subnet routing, Outbound traffic via NAT, Configuring strict security groups
IAM Fundamentals & Cloud Budgets	Principle of least privilege, Creating identity policies, Assuming cross-account roles, Navigating the AWS Billing console, Setting automated budget thresholds

Module 4

Containerization & CI/CD

Modern deployment relies on consistency. This module teaches you how to package applications so they run identically on any machine. You will master Docker, learn to deploy serverless containers in the cloud, and automate your entire build and release lifecycle using continuous integration pipelines.

Topics	
Docker & Docker Compose	Writing optimized Dockerfiles, Managing multi-container environments, Volume mounting and networking, Containerizing FastAPI backends
Provisioning AWS App Runner / ECS (Fargate)	Serverless container concepts, Configuring task definitions, Load balancer integration, Auto-scaling policy configuration
CI/CD pipelines with GitHub Actions	Creating workflow YAML files, Triggering automated builds, Managing GitHub secrets, Continuous deployment to AWS

Module 5

LLM Fundamentals & Prompting

Unlock the core capabilities of Large Language Models. Before building search engines, you must learn to control model output reliably. This module covers advanced prompt engineering, managing strict context windows, and forcing LLMs to output structured data and execute external function tools.

Topics	
Prompt Engineering & Context Management	Zero-shot and few-shot prompting, Chain of thought reasoning, Token calculation, Sliding window techniques, Context compression
Structured Outputs & Pydantic Validation	Enforcing strict JSON output schemas via APIs, Defining complex nested Pydantic models, Validating LLM responses natively against type hints, Handling and retrying output parsing errors gracefully, Using discriminator fields for union types
Tool Calling & Function Calling	Defining precise function schemas for LLMs, Parsing and validating tool arguments, Handling multi-tool parallel execution, Processing tool results into chat history, Managing hallucinated tool calls

Module 6

Vector Search & Core RAG

This module focuses heavily on data retrieval architecture. You will learn to chunk documents effectively, convert text into mathematical vector embeddings, and build high-accuracy Retrieval-Augmented Generation (RAG) pipelines that utilize hybrid search and reranking algorithms.

Topics	
Chunking & Embedding Models	Fixed-size and semantic chunking, Overlap optimization, Understanding vector representations, Dimensionality trade-offs, BM25 sparse matrices
Vector Databases (Pinecone, Qdrant)	Index creation, Distance metrics (Cosine, Euclidean), Metadata filtering, Cloud vector database provisioning
Vanilla RAG & Hybrid Search (RRF	End-to-end basic retrieval, Keyword-based search mechanisms, Combining dense and sparse signals, Implementing RRF algorithms
Retrieval Evaluation & Reranking	Precision and recall metrics, Cross-encoder reranking models, API integration for rerankers

Module 7

Enterprise Graph Architecture

Moving beyond flat documents, this module teaches you how to model and query complex, interconnected enterprise data. You will dive into Knowledge Graphs, learning to map relationships across disparate datasets and query them using Cypher to power advanced AI chatbot logic.

Topics	
Knowledge Graph Concepts & Data Modeling	Nodes, edges, and properties, Ontologies vs taxonomies, Transforming relational data tables to graph structures
Graph Databases (Neo4j) & Cypher	AuraDB cloud provisioning, Pattern matching syntax, Path traversal queries, Optimizing complex graph joins
Enterprise Integrations	Overview of Amazon Neptune, Mapping logistics networks, Modeling user interactions, Graph-based retrieval logic for AI chatbot

Module 8

Multimodal RAG & Vision AI

Enterprise data is rarely just clean text. This module prepares you to handle messy, real-world formats by integrating computer vision into your retrieval pipelines. You will learn to parse complex layouts natively without traditional OCR, process visual embeddings, and reason over charts.

Topics	
Document Parsing	Identifying document structures, Optical character recognition pipelines, OCR-free embedding strategies via ColPali
Vision Language Models & Visual Embeddings	Architecture of vision-language bridges, Prompting with images, Contrastive language-image pretraining (CLIP) concepts, Image-to-image similarity search
Handling Scanned PDFs, Tables & Charts	Extracting nested tabular data natively, Reasoning over complex chart visuals and infographs, Ingesting unstructured legacy enterprise PDFs, Aligning bounding boxes with text chunks, Managing multi-page visual context

Module 9

Agentic Frameworks & LangGraph

Evolve from building static pipelines to architecting autonomous, decision-making AI agents. You will study the theoretical design patterns that allow AI to plan and execute tasks, and gain hands-on experience using LangGraph to build robust multi-agent supervisor systems.

Topics	
Agentic Fundamentals & Design Patterns	Defining agency, ReAct framework loops, Plan & Execute structures, Supervisor and Router patterns
Multi-Agent Systems & LangGraph	Defining graphs and state management, Creating nodes and edges, Conditional routing logic, Compiling graphs
Parallel Execution & Subgraphs	Running agent tasks concurrently using async execution, Managing nested parent-child graph architectures, Aggregating parallel outputs with map-reduce patterns, Handling state conflicts during concurrent node execution

Module 10

Advanced Agent Orchestration

Take your agent architectures to production readiness. This module focuses on making agents reliable and secure by implementing long-term memory, human-in-the-loop safeguards, advanced error recovery loops, and secure enterprise communication protocols.

Topics	
Memory & Human-in-the-Loop Workflows	Check-pointing graph states, Semantic long-term memory retrieval, Interrupting graph execution, Requesting manual state approval
Error Handling & AWS Bedrock AgentCore	Detecting infinite ReAct loops, Self-correction prompting mechanisms, Managed agent provisioning, Integrating knowledge bases
Model Context Protocol (MCP)	Connecting agents to external tools safely, Host/Client/Server architectures, Standardizing tool access boundaries

Module 11

Legacy Systems & Integrations

Bridge the gap between modern AI applications and legacy corporate ecosystems. You will learn to integrate AI workflows into daily corporate tools (Slack, Jira) while simultaneously navigating older, on-premise architectures utilizing SOAP APIs and SQL servers.

Topics	
Connecting to Modern Workspaces	Webhooks and bot tokens for Slack/Teams, Interactive message payloads, Reading internal Jira pages, Automating ticket creation
SOAP APIs & XML Parsing	Understanding WSDL document structures and types, Constructing valid XML SOAP envelopes using Python libraries like Zeep, Parsing complex legacy XML responses safely, Handling SOAP faults and legacy error codes, Converting XML payloads to modern JSON formats
Oracle & MS SQL Server Integration	Establishing secure connections using Python-native drivers (pyodbc, oracledb) and SQLAlchemy, Constructing safe parameterized queries to prevent SQL injection, Implementing read-only database roles, Mapping complex database schemas to LLM context, Handling Text-to-SQL logic constraints and fallbacks

Module 12

Identity & Access Management

Security is paramount in enterprise AI. A major focus is placed on enterprise identity management, teaching you how to implement Single Sign-On (SSO) and Role-Based Access Control (RBAC) to ensure strict data governance and regulatory compliance.

Topics	
Enterprise SSO (OAuth 2.0, SAML, OIDC)	Authentication vs authorization, Understanding JWT tokens, OAuth grant types, SAML assertions
Active Directory & RBAC	Mapping Azure AD groups, Implementing Role-Based Access Control, Enforcing data-level permissions in retrieval layers

Module 13

Production AI Security & Guardrails

Focusing on the operational realities of AI systems, this module prepares you to secure deployments at scale. You will study defensive strategies against malicious prompts, implement automated data redaction, and deploy programmable guardrails to restrict model behavior.

Topics	
OWASP LLM Top 10 & Security	Identifying major LLM vulnerabilities, Prompt injection defenses, Jailbreak prevention techniques
PII Detection & Masking	Implementing Microsoft Presidio analyzers and anonymizers, Redacting sensitive entities (SSN, credit cards, emails), Customizing regex patterns for domain-specific PII, Reversing masks safely post-generation, Evaluating false-positive redaction rates
Guardrails (NeMo / Bedrock)	Writing programmable conversational rails using Colang, Configuring strict input and output filtering pipelines natively in Python, Enforcing topical boundaries to prevent off-topic chatter, Setting up AWS Bedrock managed guardrail configurations via Boto3, Testing rails against jailbreak libraries

Module 14

AI Observability & Gateway Management

Learn to manage, monitor, and evaluate your AI traffic. You will manage API traffic through unified gateways, ensure system reliability with retry logic, and establish deep system observability to trace every token and tool call in production.

Topics	
Production Reliability & LLM Gateways	Idempotency keys for safe tool execution, Exponential backoff strategies, Centralizing provider API keys via LiteLLM/Portkey, Configuring rate limiting and fallback routing
Evaluation (DeepEval, RAGAS)	Automating LLM-as-a-judge scoring pipelines, Calculating RAGAS faithfulness and answer relevance metrics, Measuring context precision and recall, Creating synthetic benchmark datasets from source documents, Tracking evaluation scores across deployment runs
Observability (LangSmith, Langfuse)	Capturing deep span-level execution traces, Monitoring granular token costs and endpoint latency, Debugging multi-step agent reasoning and tool inputs, Creating custom user-session tracking metrics, Exporting trace data for continuous model improvement

Capstone 1

OmniGuard : Secure AI Integration

In this capstone, you will experience the complete lifecycle of a technical engagement by building the OmniGuard Enterprise Connector. Acting as a Forward Deployed Engineer, you will conduct discovery workshops, draft architecture documents, and execute a complex technical build for a corporate client. The project features strict Role-Based Access Control (RBAC), secure SQL database integrations, and programmable security guardrails.

Topics	
Pillar 1 (Consulting Lifecycle)	Conducting technical discovery and scoping workshops, Defining data classifications, Drafting architecture SOWs, Preparing ROI presentations for CISO/executives, Delivering User Acceptance Testing (UAT) runbooks
Pillar 2 (Technical Build)	Executing mock OAuth 2.0 / RBAC flows, Constructing Hybrid RAG alongside secure Text-to-SQL for MS SQL databases, Implementing NeMo & Presidio guardrails, Finalizing Dockerized FastAPI cloud deployments

Capstone 2

AuditMesh : Multi-Agent Compliance System

Your final capstone tests your ability to orchestrate and secure complex, autonomous agentic workflows. You will design and deploy AuditMesh, a multi-agent supervisor system that handles financial compliance auditing. This project emphasizes human-in-the-loop approvals, secure tool execution via the Model Context Protocol (MCP), and deep token cost observability to prove enterprise readiness.

Topics	
Pillar 1 (Consulting Lifecycle)	Mapping 5-step manual compliance workflows, Identifying Human-in-the-Loop bottlenecks, Drafting latency/cost SLAs, Defining Model Context Protocol trust boundaries, Leading operations and training handoffs
Pillar 2 (Technical Build)	Developing a LangGraph Multi-Agent Supervisor, Deploying a custom MCP server for secure Jira ticketing, Building comprehensive token cost/trace dashboards, Creating Streamlit/Gradio UIs for human approval workflows